
Math 21

Algebra and Its Functions

Math 21 Course at a Glance

Algebra and Its Functions

Review Content		Functions and Function Interpretation		Polynomials and Factoring		Rational Expressions and Equations	
REVIEW		REQUIRED		REQUIRED		REQUIRED	
M21-REV-001	I can determine and interpret the slope of a line.	M21-FUN-001	I can evaluate a function from a formula, graph, or table.	M21-POL-001	I can add and subtract polynomials.	M21-RAT-001	I can state excluded values for a rational expression.
M21-REV-002	I can graph a line from an equation and write an equation from given information.	M21-FUN-002	I can determine whether a relation is a function.	M21-POL-002	I can multiply polynomials.	M21-RAT-002	I can simplify a rational expression while preserving its restrictions.
M21-REV-003	I can apply a linear function to a context.	M21-FUN-003	I can determine domain and range from a graph or table.	M21-POL-003	I can factor out a greatest common factor.	M21-RAT-003	I can multiply and divide rational expressions.
M21-REV-004	I can identify the solution of a linear system as an intersection point.	M21-FUN-004	I can solve for an input that produces a specified output.	M21-POL-004	I can factor a quadratic trinomial.	M21-RAT-004	I can add and subtract rational expressions using a common denominator.
M21-REV-005	I can solve a simple two-variable linear system using an appropriate method.	M21-FUN-005	I can interpret inputs, outputs, domain, and range in context.	M21-POL-005	I can use factoring to determine the zeros of a polynomial.	M21-RAT-005	I can solve a rational equation and reject extraneous solutions.
M21-REV-006	I can simplify an expression using basic positive-integer exponent rules.	M21-FUN-006	I can calculate and interpret average rate of change.				
		M21-FUN-007	I can compare functions represented by formulas, graphs, or tables.				

Math 21 Course at a Glance

Algebra and Its Functions

Exponents and Radicals

REQUIRED

M21-EXP-001	I can simplify expressions containing negative exponents.
M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms.
M21-EXP-003	I can simplify radical expressions.
M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents.
M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

M21-QUAD-001	I can identify key features of a quadratic function from its graph or equation.
M21-QUAD-002	I can connect standard, factored, and vertex forms to features of a parabola.
M21-QUAD-003	I can solve a quadratic equation by factoring.
M21-QUAD-004	I can solve a quadratic equation using the square-root property.
M21-QUAD-005	I can solve a quadratic equation by completing the square.
M21-QUAD-006	I can solve a quadratic equation using the quadratic formula.
M21-QUAD-007	I can choose an efficient method for solving a quadratic equation.
M21-QUAD-008	I can graph a quadratic function using its vertex, intercepts, and symmetry.
M21-QUAD-009	I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

M21-EXT-001	I can perform basic arithmetic with complex numbers.
M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers.

Math 21 In-Depth Curriculum

Algebra and Its Functions

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Review Content

REVIEW

M21-REV-001

I can determine and interpret the slope of a line.

SUPPORTING SKILLS

R	Determine slope from a graph. (first introduced in Math 12) SK-LIN-SLOPE-GRAPH · Reinforce
R	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Reinforce
R	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Reinforce

M21-REV-002

I can graph a line from an equation and write an equation from given information.

SUPPORTING SKILLS

R	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Reinforce
R	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Reinforce
R	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Reinforce
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

M21-REV-003

I can apply a linear function to a context.

SUPPORTING SKILLS

I	Explain a model parameter in the language of its context. SK-MODEL-PARAMETER-INTERPRET · Introduce
R	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Reinforce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M21-REV-004

I can identify the solution of a linear system as an intersection point.

SUPPORTING SKILLS

R	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Reinforce
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M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

SUPPORTING SKILLS

R	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Reinforce
R	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Reinforce
R	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Reinforce
R	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Reinforce

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

SUPPORTING SKILLS

R	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Reinforce
R	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Reinforce
R	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Reinforce

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Algebra and Its Functions

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Functions and Function Interpretation

REQUIRED

M21-FUN-001 I can evaluate a function from a formula, graph, or table.

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Identify the input and output represented by function notation. (first introduced in Math 12) SK-FUNC-NOTATION-READ · Apply

M21-FUN-002 I can determine whether a relation is a function.

SUPPORTING SKILLS

D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Deepen

M21-FUN-003 I can determine domain and range from a graph or table.

SUPPORTING SKILLS

D	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Deepen
D	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Deepen

M21-FUN-004 I can solve for an input that produces a specified output.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Solve for an input associated with a specified output. SK-FUNC-SOLVE-INPUT · Introduce

M21-FUN-005 I can interpret inputs, outputs, domain, and range in context.

SUPPORTING SKILLS

D	Restrict domain to values meaningful in a context. (first introduced in Math 12) SK-FUNC-CONTEXT-DOMAIN · Deepen
D	Interpret the range of a function in a context. (first introduced in Math 12) SK-FUNC-CONTEXT-RANGE · Deepen
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Deepen

M21-FUN-006 I can calculate and interpret average rate of change.

SUPPORTING SKILLS

I	Calculate average rate of change over an interval. SK-FUNC-AROC · Introduce
I	Interpret average rate of change with contextual units. SK-FUNC-AROC-INTERPRET · Introduce
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

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Algebra and Its Functions

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M21-FUN-007	I can compare functions represented by formulas, graphs, or tables.
SUPPORTING SKILLS	
I	Compare functions shown in different representations. SK-FUNC-COMPARE-REP · Introduce
I	Connect parameters in a formula to features of its graph. SK-REP-FORMULA-GRAPH · Introduce
I	Use tabular patterns to predict graph shape and behavior. SK-REP-TABLE-GRAPH · Introduce

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Algebra and Its Functions

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Polynomials and Factoring

REQUIRED

M21-POL-001

I can add and subtract polynomials.

SUPPORTING SKILLS

A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply
D	Identify like polynomial terms by variable part and exponent. (first introduced in Math 12) SK-POLY-LIKE-TERMS · Deepen
R	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Reinforce

M21-POL-002

I can multiply polynomials.

SUPPORTING SKILLS

A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
D	Multiply two binomials using distribution. (first introduced in Math 12) SK-POLY-BINOMIAL-MULTIPLY · Deepen
D	Multiply a monomial by a polynomial. (first introduced in Math 12) SK-POLY-DISTRIBUTE · Deepen
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-003

I can factor out a greatest common factor.

SUPPORTING SKILLS

R	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Reinforce
D	Factor the greatest common monomial factor from a polynomial. (first introduced in Math 12) SK-FACTOR-GCF · Deepen
A	Determine the numerical and variable greatest common factor of polynomial terms. (first introduced in Math 12) SK-POLY-GCF · Apply
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-004

I can factor a quadratic trinomial.

SUPPORTING SKILLS

A	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Apply
R	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Reinforce
R	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Reinforce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-005

I can factor a difference of squares and other required special cases.

SUPPORTING SKILLS

A	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Apply
I	Factor a difference of two squares. SK-FACTOR-DIFF-SQUARES · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

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Algebra and Its Functions

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M21-POL-006	I can use factoring to determine the zeros of a polynomial.
SUPPORTING SKILLS	
A	Factor a difference of two squares. (first introduced in Math 21) SK-FACTOR-DIFF-SQUARES · Apply
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
I	Apply the zero-product property. SK-FACTOR-ZERO-PRODUCT · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

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Algebra and Its Functions

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Rational Expressions and Equations

REQUIRED

M21-RAT-001

I can state excluded values for a rational expression.

SUPPORTING SKILLS

I

Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · Introduce

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

SUPPORTING SKILLS

I

Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · Introduce

I

Factor numerators and denominators completely.
SK-RAT-FACTOR · Introduce

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

M21-RAT-003

I can multiply and divide rational expressions.

SUPPORTING SKILLS

A

Add, subtract, multiply, and divide fractions. (inherited from middle school)
SK-NUM-FRACTION-OPERATE · Apply

A

Cancel common factors rather than individual terms. (first introduced in Math 21)
SK-RAT-CANCEL-FACTORS · Apply

A

Factor numerators and denominators completely. (first introduced in Math 21)
SK-RAT-FACTOR · Apply

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

SUPPORTING SKILLS

A

Add, subtract, multiply, and divide fractions. (inherited from middle school)
SK-NUM-FRACTION-OPERATE · Apply

I

Combine rational expressions over a common denominator.
SK-RAT-COMBINE · Introduce

I

Determine the least common denominator of rational expressions.
SK-RAT-LCD · Introduce

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

I

Clear denominators from a rational equation.
SK-RAT-CLEAR-DENOMINATORS · Introduce

I

Check candidate solutions against denominator restrictions.
SK-RAT-EXTRANEIOUS · Introduce

A

Determine excluded values from an original denominator. (first introduced in Math 21)
SK-RAT-RESTRICTIONS · Apply

Exponents and Radicals

REQUIRED

M21-EXP-001	I can simplify expressions containing negative exponents.
SUPPORTING SKILLS	
I	Rewrite a negative exponent using a reciprocal. SK-EXP-NEGATIVE · Introduce
A	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Apply
A	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Apply

M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms.
SUPPORTING SKILLS	
I	Interpret numerator and denominator roles in a rational exponent. SK-EXP-RATIONAL · Introduce
I	Convert between radical and rational-exponent notation. SK-RAD-CONVERT · Introduce

M21-EXP-003	I can simplify radical expressions.
SUPPORTING SKILLS	
A	Express a whole number as a product of prime factors. (inherited from middle school) SK-NUM-PRIME-FACTOR · Apply
I	Extract perfect-power factors from a radical. SK-RAD-FACTOR-PERFECT · Introduce

M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents.
SUPPORTING SKILLS	
A	Rewrite a negative exponent using a reciprocal. (first introduced in Math 21) SK-EXP-NEGATIVE · Apply
D	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Deepen
D	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Deepen
D	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Deepen
A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply
I	Apply the zero-exponent rule. SK-EXP-ZERO · Introduce

M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Require an even-root radicand to be nonnegative when determining domain. SK-FUNC-DOMAIN-EVEN-ROOT · Introduce
I	Check a candidate solution for extraneous results. SK-RAD-CHECK · Introduce
I	Isolate a radical expression before solving. SK-RAD-ISOLATE · Introduce

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Algebra and Its Functions

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Quadratic Functions and Equations

REQUIRED

M21-QUA D-001	I can identify key features of a quadratic function from its graph or equation.
SUPPORTING SKILLS	
I	Identify standard, factored, and vertex forms. SK-QUAD-FORM-IDENTIFY · Introduce
I	Determine x- and y-intercepts of a quadratic function. SK-QUAD-INTERCEPTS · Introduce
I	Determine the vertex and axis of symmetry. SK-QUAD-VERTEX · Introduce

M21-QUA D-002	I can connect standard, factored, and vertex forms to features of a parabola.
SUPPORTING SKILLS	
I	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. SK-QUAD-FORM-CONVERT · Introduce
D	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Deepen
A	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Apply
A	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M21-QUA D-003	I can solve a quadratic equation by factoring.
SUPPORTING SKILLS	
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Deepen
I	Determine real and complex zeros using an appropriate method. SK-POLY-ZEROS · Introduce

M21-QUA D-004	I can solve a quadratic equation using the square-root property.
SUPPORTING SKILLS	
I	Apply the square-root property with both signs. SK-QUAD-SQUARE-ROOT · Introduce
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply

M21-QUA D-005	I can solve a quadratic equation by completing the square.
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Create a perfect-square trinomial by completing the square. SK-QUAD-COMPLETE-SQUARE · Introduce

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Algebra and Its Functions

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M21-QUA D-006	I can solve a quadratic equation using the quadratic formula.
SUPPORTING SKILLS	
I	Use the discriminant to predict the number and type of solutions. SK-QUAD-DISCRIMINANT · Introduce
I	Substitute coefficients accurately into the quadratic formula. SK-QUAD-FORMULA · Introduce
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply

M21-QUA D-007	I can choose an efficient method for solving a quadratic equation.
SUPPORTING SKILLS	
A	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Apply
A	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Apply
A	Substitute coefficients accurately into the quadratic formula. (first introduced in Math 21) SK-QUAD-FORMULA · Apply
I	Select an efficient quadratic-solving method from equation structure. SK-QUAD-METHOD-SELECT · Introduce
A	Apply the square-root property with both signs. (first introduced in Math 21) SK-QUAD-SQUARE-ROOT · Apply

M21-QUA D-008	I can graph a quadratic function using its vertex, intercepts, and symmetry.
SUPPORTING SKILLS	
A	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M21-QUA D-009	I can create and interpret a quadratic model for a projectile, area, or optimization problem.
SUPPORTING SKILLS	
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
I	Interpret vertex and zeros in an application. SK-QUAD-MODEL-FEATURES · Introduce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Extension Content

EXTENSION

M21-EXT-001	I can perform basic arithmetic with complex numbers.
SUPPORTING SKILLS	
I	Simplify integer powers of the imaginary unit. SK-CPLX-I-POWERS · Introduce
I	Add, subtract, and multiply complex numbers. SK-CPLX-OPERATE · Introduce

M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers.
SUPPORTING SKILLS	
A	Simplify integer powers of the imaginary unit. (first introduced in Math 21) SK-CPLX-I-POWERS · Apply
I	Express a quadratic solution with a negative discriminant in complex form. SK-CPLX-QUAD-SOLUTION · Introduce
A	Use the discriminant to predict the number and type of solutions. (first introduced in Math 21) SK-QUAD-DISCRIMINANT · Apply